

# Lab Class: Physics

## Real Gas Experiment

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## 1 Introduction

This experiment is part of a university course and serves as a learning tool. Unlike ideal gases, real gases have a complex phase transition. This experiment shows the behavior of the gas  $SF_6$  (sulfur hexafluoride) near its critical point, using pressure, volume and temperature relationships.

## 2 Theory

The general gas equation for an ideal gas is stated as:

$$pV_m = RT \quad (1)$$

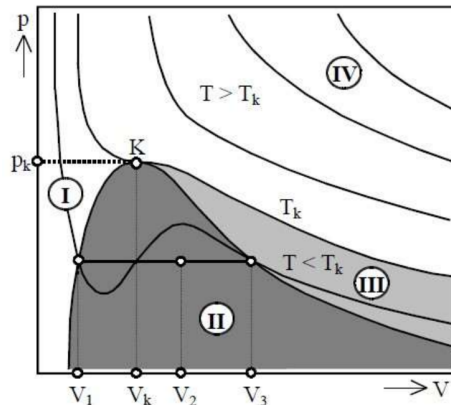
where  $p$  is pressure,  $V_m$  is the molar volume of a gas,  $R$  is the ideal gas constant ( $8.314 \frac{J}{molK}$ ) and  $T$  is the temperature.

The Van der Waals equation modifies the ideal gas equation to account for molecular volume and attractive forces between the molecules. With this modification the real gas equation is formed and can be read as:

$$\left(p + \frac{a}{V_m^2}\right) \cdot (V_m - b) = RT \quad (2)$$

where  $a$  is the intermolecular attraction parameter and  $b$  is the excluded volume parameter. They can be determined using the critical point.

A phase transition occurs when a substance changes from one state of matter to another, due to variations in temperature and pressure. For real gases this can be observed using a pressure-volume ( $pV$ ) diagram with constant temperatures. The critical point of a real gas is the endpoint of the liquid-gas coexistence curve. While plotting a  $pV$  diagram, four areas of phases occur.



(3)

**Area I:** Liquid state.

**Area II:** Gas-liquid transition ("wet vapor").

**Area III:** A gas that can be liquefied by isothermal compression.

**Area IV:** A gas that can't be liquefied by isothermal compression.

Point *K* indicates the critical point.

*Vapor pressure* is the pressure exerted by a real gas in equilibrium with its liquid (or solid) phase at a given temperature. It represents the tendency of molecules to escape from the liquid into the gas phase. The energy for this transformation is known as *Enthalpy of vaporization*. For this to occur, the temperature should be much lower than the critical temperature. This is because once the gas reaches its critical temperature, it no longer has a phase shift. Vapor pressure can very well be defined using the Clausius-Clapeyron equation. Which is stated as:

$$\frac{dp}{dT} = \frac{H_{vap}}{T \cdot V_{gas}} \quad (4)$$

where  $H_{vap}$  is the enthalpy of vaporization. This equation holds true whenever the temperature is far below the critical temperature where the volume of the gas is much more than the volume of the liquid form. For this experiment the difference quotient  $\frac{\Delta p}{\Delta T}$  was used instead of the differential quotient  $\frac{dp}{dT}$ .

By the assumption of the gas to be ideal:

$$V_{gas} = \frac{nRT}{p}$$

In reference to these conditions, the *molar enthalpy of vaporization* and the *specific enthalpy of vaporization* of a gas are given by:

$$H_{vap,m} = \frac{T^2 \cdot R \cdot \Delta p}{\Delta T \cdot p} \quad (5)$$

( $H_{vap,m} \hat{=}$  molar enthalpy of vaporization)

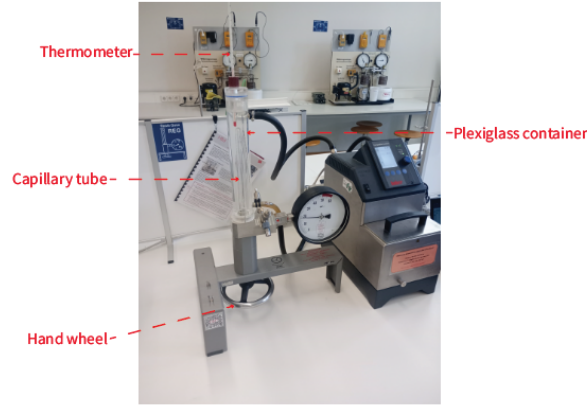
$$h_{vap} = \frac{T^2 \cdot R_s \cdot \Delta p}{\Delta T \cdot p} \quad (6)$$

( $h_{vap} \hat{=}$  specific enthalpy of vaporization), where  $nR = mR_s$ .

### 3 Experimental Setup

The experiment was set up in a laboratory provided by the *Hochschule München University of Applied Sciences*, which is used for a variety of experiments on thermodynamics.

For this experiment an enclosed capillary tube with liquid mercury and sulfur hexafluoride ( $SF_6$ ) gas was used. The function of the mercury is to apply pressure on the gas. The pressure can be adjusted with a hand wheel which controls the amount of mercury in the tube. The tube is surrounded by a transparent Plexiglass container filled with water. The water bath ensures a constant temperature in the capillary tube and therefore of the  $SF_6$  gas in it.



Manometer



Thermostat

(7)

The capillary is scaled from 0 to  $4\text{ cm}^3$  which gradually decreases by  $0.05\text{ cm}^3$  from bottom to top. A thermometer inserted through an opening at the top of the container was used to measure the temperature. The water is supplied from a circulating thermostat which can be used to alter the temperature within the system. The thermostat can be set to temperatures between  $10^\circ\text{C}$  and  $45^\circ\text{C}$  using a rotary knob or wheel. The main purpose of the pressure chamber was to adjust the volume ratio of the  $\text{SF}_6$  gas and the mercury. This was done by rotating the hand wheel left and right. The current pressure on the gas in the tube is displayed in  $\text{bar}$  ( $10^5\text{ Pa}$ ) with a large manometer.

## 4 Experimental Procedure

The experiment was carried out by adjusting the volume of the sulfur hexafluoride ( $\text{SF}_6$ ) in the capillary tube. Initially the volume was set to be relatively large ( $3.6\text{ cm}^3$ ) and then reduced, first by steps of  $0.2\text{ cm}^3$  (for  $V > 2\text{ cm}^3$ ) then by steps of  $0.1\text{ cm}^3$ . The pressure measurement for each volume was recorded.

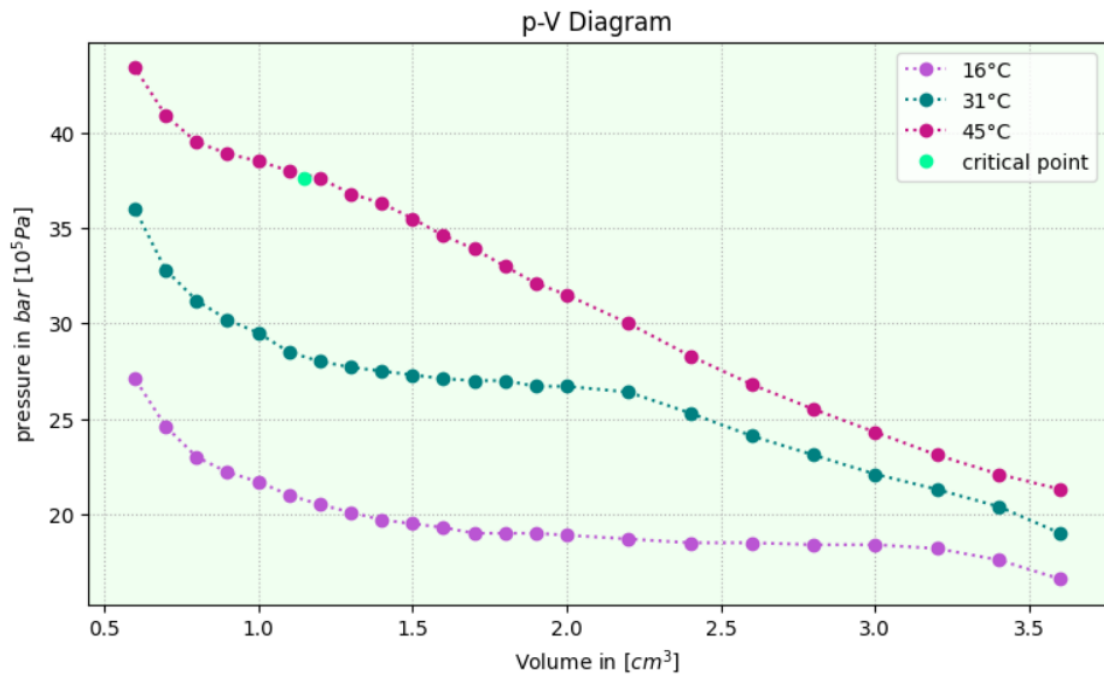
This was done for three different temperatures ( $16^\circ\text{C}$ ,  $31^\circ\text{C}$  and  $45^\circ\text{C}$ ). For the temperatures  $16^\circ\text{C}$  ( $289.15\text{ K}$ ) and  $31^\circ\text{C}$  ( $304.15\text{ K}$ ), it was possible to record the pressure for down to a volume of  $0.5\text{ cm}^3$ . But for the temperature  $45^\circ\text{C}$  ( $318.15\text{ K}$ ), the pressure read above  $50\text{ bar}$  at the volume of  $0.5\text{ cm}^3$ , which was a red zone in the manometer. So this point was neglected for all three measurements. The measured values are as follows:

| V of SF <sub>6</sub> (in cm <sup>3</sup> ) | p at 16°C(in bar) | p at 31°C(in bar) | p at 45°C(in bar) |
|--------------------------------------------|-------------------|-------------------|-------------------|
| 3.6                                        | 16.6              | 19                | 21.3              |
| 3.4                                        | 17.6              | 20.4              | 22.1              |
| 3.2                                        | 18.2              | 21.3              | 23.1              |
| 3                                          | 18.4              | 22.1              | 24.3              |
| 2.8                                        | 18.4              | 23.1              | 25.5              |
| 2.6                                        | 18.5              | 24.1              | 26.8              |
| 2.4                                        | 18.5              | 25.3              | 28.3              |
| 2.2                                        | 18.7              | 26.4              | 30                |
| 2                                          | 18.9              | 26.7              | 31.5              |
| 1.9                                        | 19                | 26.7              | 32.1              |
| 1.8                                        | 19                | 27                | 33                |
| 1.7                                        | 19                | 27                | 33.9              |
| 1.6                                        | 19.3              | 27.1              | 34.6              |
| 1.5                                        | 19.5              | 27.3              | 35.5              |
| 1.4                                        | 19.7              | 27.5              | 36.3              |
| 1.3                                        | 20.1              | 27.7              | 36.8              |
| 1.2                                        | 20.5              | 28                | 37.6              |
| 1.1                                        | 21                | 28.5              | 38                |
| 1                                          | 21.7              | 29.5              | 38.5              |
| 0.9                                        | 22.2              | 30.2              | 38.9              |
| 0.8                                        | 23                | 31.2              | 39.5              |
| 0.7                                        | 24.6              | 32.8              | 40.9              |
| 0.6                                        | 27.1              | 36                | 43.4              |

(8)

#### a. pV diagram

Based on the recorded values, it was possible to draw the pV diagram of the three isotherms:



(9)

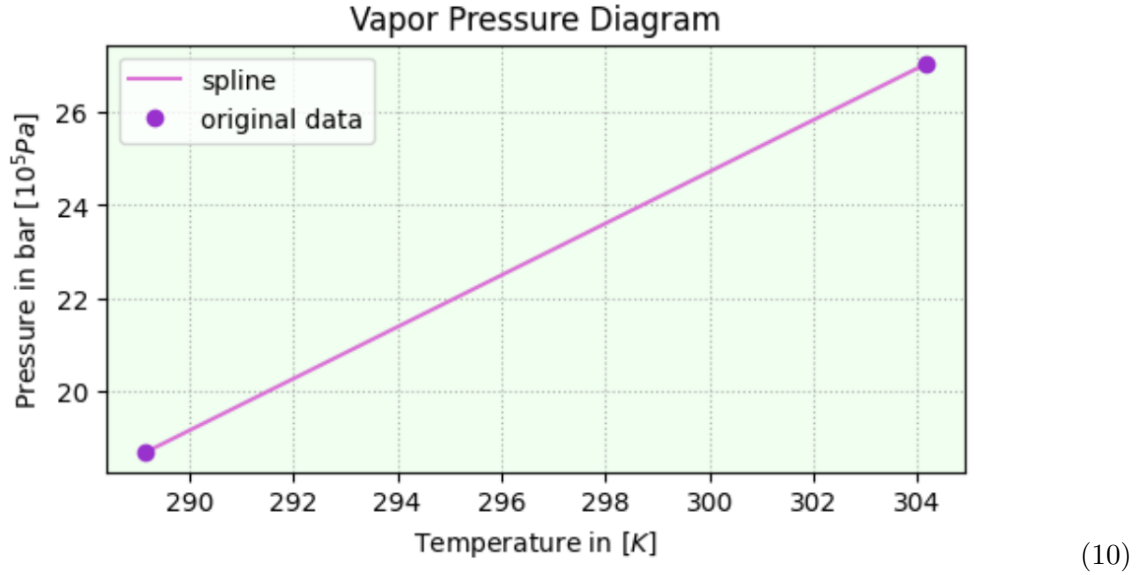
It must be made clear that the lines connecting the data points are interpolated solely for better readability and not backed up by data or calculation.

From the graph, an almost horizontal flatness of the curves can be observed at a certain pressure. As 45°C is very close to the critical temperature of SF<sub>6</sub> gas (45.5°C), this flatness is almost non-existent. These constant pressures at the different temperatures are called the **vapor pressure**

of the gas at that temperature. Vapor pressure of a gas is defined as the pressure exerted by the gas when it's in equilibrium with its liquid or solid form in a closed container. By analyzing the graph it was possible to state the vapor pressure  $\approx 18.66\text{bar}$  for  $16^\circ\text{C}$  and  $27.04\text{bar}$  for  $31^\circ\text{C}$ . The critical point was observed to be at a pressure of  $37.5\text{bar}$  and a volume of  $1.15\text{cm}^3$ .

### b. Vapor pressure vs T graph

From the information obtained in the graph above, the vapor pressure as a function of temperature was plotted:



Since it is not possible to determine the vapor pressure at  $45^\circ\text{C}$ , due to its proximity to the critical temperature, only the vapor pressures at the two lower temperatures was used. This gave an expected straight line.

### c. Molar/Specific enthalpy of vaporization

Based on the data measured and determined, the values of molar enthalpy of vaporization and specific enthalpy of vaporization were calculated. This was done using the equations (5) and (6) from the theory:

**Molar enthalpy of vaporization:**  $H_{vap,m} = 21.9123 \frac{\text{kJ}}{\text{mol}}$

**Specific enthalpy of vaporization:**  $h_{vap} = 122.5053 \frac{\text{kJ}}{\text{kg}}$

These values were calculated under the following assumptions:

$T$  was the average temperature of  $16^\circ\text{C}$  and  $31^\circ\text{C}$  with a value of  $296.65\text{K}$  ( $23.5^\circ\text{C}$ ).

$p$  was the average vapor pressure of  $18.66\text{bar}$  and  $27.04\text{bar}$  with a value of  $22.8514\text{bar}$  ( $2.2851\text{MPa}$ ).

$\Delta p$  was the difference between the two vapor pressures ( $27.04285\text{bar} - 18.66\text{bar} = 8.38285\text{bar} = 838.285\text{kPa}$ ).

$\Delta T$  was the difference between the two temperatures ( $304.15\text{K} - 289.15\text{K} = 15\text{K}$ ).

The specific gas constant  $R_s$  was obtained by dividing the ideal gas constant  $R$  by the molar mass of  $\text{SF}_6$ .  $R_s = \frac{R}{M_{\text{SF}_6}} = \frac{8.314 \frac{\text{J}}{\text{mol}\cdot\text{K}}}{146.06 \frac{\text{g}}{\text{mol}}} = 56.9218 \frac{\text{J}}{\text{kg}\cdot\text{K}}$ .

## 5 Analysis

During the measurements for the two lower temperatures, a gradually increasing amount of gas was liquidized. The start of this process could be observed at  $2.8\text{cm}^3$  for the  $16^\circ\text{C}$  measurements and at  $2\text{cm}^3$  for the  $31^\circ\text{C}$  measurements. For the measurements at  $45^\circ\text{C}$  this was barely seen, since it was too close to the critical point, therefore no clear separation between gas and liquid can be defined.

Another observation could be made, when setting the pressure from  $0.5\text{cm}^3$  back to  $3.6\text{cm}^3$  to obtain the measurements for another temperature. While the volume was increased again, the liquid was boiling back to its initial gaseous form.

This behavior supports the data displayed in the pV diagram in figure (9), where the transition of gas to liquid is represented by the flat part of the curves.

It was clearly observed that the pressure decreased with the increase of volume, hence supporting the theory. The recorded critical pressure ( $37.5\text{bar}$ ) was a very close point to the literature value ( $37.6\text{bar}$ ).

Usually a vapor pressure-temperature graph has an exponential shape. In this experiment however, the data forms a linearly increasing graph (figure (10)), due to the lack of more than two vapor pressure values.

The literature value for the molar enthalpy of vaporization for  $\text{SF}_6$  is approximately  $23.2\frac{\text{kJ}}{\text{mol}}$  which is slightly higher than the value calculated above. The reasons for this deviation could be an insufficient number of measurements, the usage of the difference quotient  $\frac{\Delta p}{\Delta T}$  instead of the differential quotient  $\frac{dp}{dT}$  and human imprecision while reading the values for volume and pressure. The literature value for the specific enthalpy of vaporization for  $\text{SF}_6$  is approximately  $138.3\frac{\text{kJ}}{\text{kg}}$ , which is close to the value achieved by this experiment. The same reasons for deviation can be assumed here.

## 6 Conclusion

Based on the findings and analysis of this experiment, it can serve to show the characteristic of a real gas. It was observed that the characteristic of the  $\text{SF}_6$  gas around the critical temperature is closely related to that of an ideal gas. The critical point of this gas was also experimentally confirmed.

### Literature

*NIST Chemistry WebBook* ( $\text{SF}_6$  properties)

Van der Waals, J. D. (1873), *On the Continuity of the Gaseous and Liquid States*

*IEC 60376* (International Electrotechnical Commission)